Does Recognition Drive Reform? School-Level Competition and Equitable eLMS Adoption in Tanzania

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1. Introduction

1.1 Background

Tanzania faces one of the most acute foundational learning crises in sub-Saharan Africa. National and cross-country assessments consistently find that the majority of primary school children cannot read a simple passage or perform basic arithmetic by the end of Standard 2, despite near-universal enrollment (Uwezo Tanzania, 2019; PAL Network, 2025). Teacher quality has been identified as the binding constraint: approximately 80 percent of Tanzania's 200,000 public school teachers had received no formal professional development in the five years prior to 2020, and real per-child public expenditure has declined as demographic pressure has outpaced fiscal capacity (Cilliers, 2025; Todd et al., 2025).

In response, the Government of Tanzania launched the National Framework for Teacher Continuous Professional Development in 2020, implemented under the Swahili acronym MEWAKA (Mafunzo Endelevu ya Walimu Kazini). MEWAKA is one of the most ambitious teacher reform programmes in the region, now operating in more than half of Tanzania's 184 districts and supported by a $500 million World Bank BOOST programme, the Bill and Melinda Gates Foundation, and the What Works Hub for Global Education (WWHGE). The programme has two interlocking components: school-based Communities of Learning (CoLs), where teachers gather weekly in peer-led professional development sessions; and an electronic Learning Management System (eLMS), developed by the Tanzania Institute of Education (TIE) with the University of Dar es Salaam, which hosts self-paced modules and CoL content that teachers can access independently.

The eLMS is central to MEWAKA's vision of scalable, low-cost teacher development. Since its launch in 2022, over 112,000 teachers have registered on the platform. Yet administrative data reveal a stark engagement gap: most registered teachers log in only once, average time since last login exceeds one year, and fewer than 2 percent have completed a module assessment (Tanzania Institute of Education, 2025).

1.2 Motivation

Low eLMS engagement is not primarily a capacity or access problem. Most non-engaging teachers are registered users who have logged in at least once, confirming platform access (Tanzania Institute of Education, 2025). Our hypothesis is that engagement is entirely invisible above the level of the individual teacher: completion carries no formal consequence and signals nothing to headteachers or WEOs. This is a stakes problem.

The accountability literature identifies a low-cost solution. Figlio and Loeb (2011) establish that making school performance visible to middle-tier administrators — without formal sanctions — can activate reputational pressure sufficient to drive behavioral responses. Cilliers, Mbiti, and Zeitlin (2021) demonstrate this mechanism in Tanzania: Big Results Now's (BRN) within-district school rankings raised learning outcomes by approximately 0.25 standard deviations for bottom-decile schools, operating entirely through DEO and headteacher reputational incentives rather than parental responses or financial consequences. The same study, however, documents a warning: ranked schools excluded low-performing students from exams, inflating average scores at the cost of the most vulnerable. This finding motivates our equity research question and our choice of a metric — eLMS completion — that cannot be gamed by student exclusion.

Habyarimana and Cilliers (2024) provide the most direct precedent. In a randomized study of school governance reforms in Tanzania, they find that WEO-targeted text messages summarizing school inspection reports — signed by supervisors and officially endorsed — improved teaching practices by 0.27 standard deviations and student learning by 0.11 standard deviations, at negligible cost. Without the messages, the programme produced no impact: the weak link was not programme quality but the failure of information to reach the middle-tier actor with the mandate and authority to act. The present study applies this logic to MEWAKA: WEOs currently receive

no information about school-level eLMS engagement despite conducting termly school visits and supervising headteachers. A monthly ward-level engagement report fixes this information gap without requiring institutional reform.

The equity problem with competition designs

Figlio and Loeb (2011) and the BRN experience both warn that ranking interventions risk concentrating responses among schools already positioned to respond, leaving weaker schools behind. Rural, low-resource schools in Tanzania face greater connectivity constraints and weaker administrative capacity. A recognition intervention that raises average engagement while widening this gap would be inequitable. The interaction between treatment assignment and school resource level is therefore pre-specified as a primary hypothesis, on equal footing with the average treatment effect.

This study proposes a randomized evaluation of a ward-level public recognition intervention designed to increase school-level eLMS engagement and to test whether its effects are equitable across school resource levels. Wards within the sampled frame are randomly assigned to receive — or not receive — a monthly ward-level eLMS engagement report, which is then shared with all schools in the ward through the WEO. We measure not only whether the intervention raises average engagement, but whether it does so equally for high- and low-resource schools, or whether it exacerbates existing inequalities. Alongside eLMS administrative outcomes, a post-intervention survey characterises which teachers within treated schools respond most strongly — generating evidence on who the emergent 'power users' of the reform are, and what individual and school-level characteristics predict uptake.

The study is embedded within the existing WWHGE Tanzania Education Lab (EdLab) operated by gui2de at Georgetown University in partnership with TIE, the University of Dar-es-Salaam, and the University of Delaware. The intervention uses data infrastructure that already exists, targets a decision that TIE can implement at negligible cost, and generates evidence directly actionable by the programme's government owner.

1.3 Research Questions

This study addresses three questions.[1]

- First: does making school-level eLMS engagement visible to WEOs through a monthly ward-level ranking increase module completion rates relative to the status quo?
- Second: are the effects of this recognition intervention equitably distributed across schools with different resource levels, or does public competition exacerbate pre-existing gaps?
- Third: within treated schools, what teacher and school characteristics predict sustained engagement — and what does this reveal about who the reform is currently reaching and failing to reach?

[1] Please note that this study does not attempt to estimate the effect of eLMS engagement on student learning outcomes. The causal chain from platform engagement to classroom practice change to measured learning gains requires a longer study window and assessment infrastructure beyond the scope of this proposal. Student learning is the ultimate motivation for this work and is articulated in the Theory of Change; it is not a measured outcome in this study.

2. Experimental Design

2.1 Intervention and Manipulation Studied

2.1.1 Present Status

Under the current MEWAKA rollout, eLMS engagement is entirely invisible above the level of the individual teacher. A teacher who completes a module receives an on-platform digital certificate; nothing is communicated to their headteacher, their WEO, or any other actor in the administrative hierarchy. Completion carries no formal consequence and confers no social signal beyond the platform itself. In a bureaucratic environment where teachers respond to hierarchical oversight and where career norms are shaped by the signals sent by supervisors, this invisibility is a design failure.

2.1.2 The Intervention: ward-level public recognition

The intervention is a monthly Ward-Level eLMS Engagement Report delivered to Ward Education Officers (WEOs) in treated wards. The report is a one-page document — produced automatically from TIE's existing Matomo analytics dashboard — ranking all schools in the ward by the share of registered teachers who completed at least one eLMS module in the preceding month. It is framed explicitly as a performance ranking, not a monitoring tool: the language is positive and celebratory, recognising the top-performing schools rather than flagging low performers.

The report is delivered to the WEO via SMS link and, where feasible, printed and posted at the ward education office. WEOs are not given any scripted instructions for how to use the information — the intervention tests whether mere visibility to a supervisory actor, with no additional accountability mechanism, is sufficient to shift school-level engagement. This is the minimal-cost version of the design. A second treatment arm, described below, adds a structured accountability step.

Why the WEO is the right recipient

WEOs sit at the most operationally consequential point in Tanzania's education hierarchy. They conduct termly school visits, sign off on CoL attendance reports, and are the first-line supervisor for headteachers. Unlike district or regional officers, WEOs have direct, repeated contact with individual schools and are known personally to headteachers and teachers. Making eLMS performance visible to the WEO therefore activates an accountability relationship that already exists, rather than creating a new one. Critically, teachers do not need to know the WEO is receiving the report for the effect to operate — headteachers, aware of the WEO's new information, may themselves create internal school-level pressure.

2.1.3 Treatment arms

The study uses a three-arm design, randomized at the ward level:

Arm	Description	Mechanism tested
Control (C)	Standard MEWAKA rollout. No engagement report shared with WEO. Status quo.	Baseline — no stakes, no visibility

Arm	Description	Mechanism tested
Treatment 1 — Visibility (T1)	Monthly ward-level school ranking delivered to WEO. No additional instructions given. The report is positive in framing, celebrating top schools.	Does visibility to a supervisory actor alone shift engagement, without explicit accountability pressure?
Treatment 2 — Visibility + Accountability (T2)	Same monthly ranking as T1, plus a structured prompt to the WEO to discuss the report with the bottom two schools during their next scheduled visit. WEO records a brief note on the discussion.	Does adding a light accountability step — a prompted conversation, not sanctions — produce additional effects beyond visibility alone?

If T1 produces effects comparable to T2, the mechanism is social visibility — teachers and headteachers respond to being seen, even without explicit accountability pressure. If T2 substantially outperforms T1, the mechanism requires active supervisory engagement, which has different cost and scalability implications. Both are policy-relevant findings. [2]

2.2 Enrolment Criteria and Sampling Strategy

2.2.1 Unit of randomization and unit of observation

The unit of randomization is the ward. The unit of observation is the school (for primary outcomes) and the individual teacher (for secondary and heterogeneity outcomes). Randomizing at the ward level is necessary because the intervention is delivered to the WEO, who covers all schools in their ward — school-level randomization within wards would contaminate the control group through the WEO's awareness of the report.

Wards in Tanzania are administrative units sitting below the district and above the school. Each ward typically contains between 3 and 8 primary schools and is served by one WEO. The ward structure is therefore a natural cluster for the intervention and aligns with the administrative hierarchy through which MEWAKA is implemented.

2.2.2 Eligibility criteria

Wards are eligible for inclusion if they meet all of the following criteria:

- Located in a district where MEWAKA has been active for at least one full academic year, ensuring baseline eLMS registration exists among teachers in the ward's schools.
- At least 20 teachers across the ward's schools are registered on the eLMS, as recorded in TIE administrative data — ensuring a sufficient base of registered users for the engagement outcome to be meaningful. A minimum absolute threshold is used in preference to a proportional threshold (e.g., 60 percent) because a proportional threshold would inadvertently restrict the sample to the most digitally advanced wards, undermining the equity research question. Low-registration wards are precisely the units where the intervention's distributional effects are most important to measure.

[2] It is important to note that the intervention does not provide additional eLMS training, devices, data bundles, or content improvements. These are held constant across all arms. Any engagement effect is therefore attributable to the recognition/visibility mechanism, not to supply-side changes.

- The ward has a functioning, identifiable WEO who is reachable by SMS — a basic implementation requirement for T1 and T2.
- The ward is not already participating in another EdLab or TIE experimental study, to avoid contamination with concurrent interventions.

Schools within eligible wards are included automatically — there is no school-level eligibility screening. This preserves the integrity of ward-level randomization and avoids selection bias at the school level. Individual teachers are the unit of observation for secondary outcomes; all teachers registered on the eLMS in sampled schools are included.

2.2.3 Sampling frame and expected sample

Sampling frame: MEWAKA is currently active in more than half of Tanzania's 184 districts (approximately 92+ districts). The EdLab documents an average of approximately 22 wards per district, yielding a rough upper bound of approximately 2,000 wards in the programme's current footprint. After applying the eligibility filters above — minimum teacher registration, functioning WEO, and no concurrent study participation — the eligible sampling frame will be smaller. The exact number will be confirmed against TIE administrative data prior to randomization; we do not pre-specify a point estimate here to avoid anchoring on an unverified assumption. The target sample of 120 wards represents approximately 6 percent of the rough upper-bound estimate, making the study feasible within the existing programme footprint even under conservative eligibility assumptions.

Target sample: Power calculations (Section 4) indicate that 40 wards per arm — 120 wards in total — is sufficient to detect a minimum detectable effect of approximately 0.23 standard deviations of the school-level completion rate, equivalent to roughly 2.3 percentage points under conservative variance assumptions, at 80 percent power and a 5 percent significance level, accounting for within-ward clustering. This MDE is calibrated to detect effects of the magnitude documented by analogous low-stakes informational interventions operating through bureaucratic reputational channels in Tanzania: Cilliers, Mbiti, and Zeitlin (2021) found effects of approximately 0.25 standard deviations from BRN school rankings, and Habyarimana and Cilliers (2024) found effects of 0.27 standard deviations on teaching practices from WEO-targeted SMS. With an average of 5 schools per ward and approximately 15 registered teachers per school, 120 wards yields a sample of roughly 600 schools and 9,000 teachers. Wards are drawn from across multiple districts: with approximately 22 wards per district and a target of 120 wards total, the study will span approximately 15–20 districts, sampling on average 6-8 wards per district (2-3 per arm). This ensures that the majority of wards in any given district remain as controls, limiting spillover through WEO-to-WEO communication within a district.

2.2.4 Stratification

Wards will be stratified prior to randomization on two dimensions:

	T1 (Visibility)		T2 (Visibility + Accountability)		Control		Total
Stratum	**High resource**	**Low Resource**	**High resource**	**Low Resource**	**High resource**	**Low Resource**	

Above median completion	10 wards	10 wards	10 wards	10 wards	10 wards	10 wards	60 wards
Below median completion	10 wards	10 wards	10 wards	10 wards	10 wards	10 wards	60 wards
Wards	20	20	20	20	20	20	**120**
Schools (~5/ward)	100	100	100	100	100	100	**600**
Teachers (~15/school)	1,500	1,500	1,500	1,500	1,500	1,500	**9,000**

Wards within each stratum are randomly assigned to C, T1, or T2 using a pre-registered randomization procedure implemented in Stata. School and teacher counts are approximate, based on an average of 5 schools per ward and 15 registered eLMS teachers per school drawn from TIE administrative records.

- **School resource level (low-resource vs. higher-resource):** Defined using a composite of four school- and ward-level indicators available from administrative sources: (i) number of registered eLMS teachers per school (from TIE records); (ii) school size by total enrolment (from TIE/EMIS records); (iii) MEWAKA implementation phase — early-phase LGAs (the original 26 from 2022/23) have received substantially more institutional support, training, and EdLab engagement than later-phase ones; and (iv) mobile broadband coverage at the ward level, drawn from GSMA or the Tanzania Communications Regulatory Authority (TCRA) coverage data. The composite is standardised and split at the median to classify wards as low-resource or higher-resource prior to randomization. This is the primary equity stratification variable — it ensures balanced representation of low-resource wards across all three treatment arms and enables the pre-specified heterogeneity test in H3.
- **Baseline eLMS engagement (median ward-level module completion rate):** Derived from Matomo analytics in the term immediately prior to randomization, aggregated to ward level. Stratifying on baseline engagement controls for pre-existing variation in platform use and improves statistical power by reducing residual outcome variance. It also ensures that the treatment and control arms are balanced on the most important predictor of the outcome variable.

2.3 Assignment Protocol and Identification Strategy

2.3.1 Randomization procedure

Randomization is conducted by the research team prior to the start of the intervention period, using a pre-registered random seed. The procedure is as follows:

- The full list of eligible wards is compiled from TIE administrative data and verified against WEO contact records.
- Wards are sorted into the four stratification cells described in Section 2.2.4.
- Within each stratum, wards are randomly ordered using a computer-generated random number, then assigned sequentially to C, T1, T2, C, T1, T2… to ensure equal allocation.
- The randomization list is locked and registered prior to any data collection or intervention delivery.

The study utilizes a 'Status Quo' counterfactual to isolate the impact of social recognition. Schools in the control wards will have full access to the eLMS platform as per the national rollout but will receive no supplementary communications, rankings, or performance nudges beyond the standard MEWAKA rollout. This pure control design ensures that any observed difference in completion rates is attributable specifically to the recognition-based intervention rather than simple information provision or platform availability.

2.3.2 Identification strategy

The identification strategy is a cluster-randomized controlled trial (cluster-RCT) with ward as the unit of randomization. Because ward assignment is determined by the research team using a pre-registered random procedure, treatment status is independent of all pre-intervention characteristics of schools, teachers, and WEOs by design. This eliminates the selection bias that would arise from any observational comparison of wards where WEOs voluntarily adopted recognition practices versus those where they did not.

The estimand of interest is the Intent-to-Treat (ITT) effect: the average difference in school-level eLMS module completion rates between treatment and control wards, regardless of the degree to which individual WEOs engage with the report. ITT is the appropriate estimand for a policy evaluation of this kind, because it captures the real-world effect of deploying the intervention at scale, where compliance will inevitably vary across WEOs. A secondary analysis will estimate the Local Average Treatment Effect (LATE) among compliant WEOs — those who open the report link, confirmed via SMS delivery tracking — using WEO compliance as an instrument for effective exposure.

One limitation of this instrument is that link-opening is a weak proxy for substantive WEO engagement with the report — a WEO may open the link without reading or acting on the content, and conversely may discuss the report with headteachers based on a printed copy without triggering the SMS link. The LATE estimate should therefore be interpreted as the effect among WEOs who at minimum received and opened the report, not among WEOs who fully processed and acted on it. Headteacher awareness of the ranking, measured in the endline survey, provides a complementary check on whether the information actually traveled down the administrative hierarchy.

2.3.3 Equity Identification and Distributional Effects

The distributional effects of the intervention are identified through two pre-specified interactions between treatment assignment and baseline ward and school characteristics. Both are primary hypotheses, not post-hoc subgroup analyses.

Resource equity interaction: Treatment is interacted with the composite resource index described in Section 2.2.4 — comprising registered teacher count, school enrolment, MEWAKA implementation phase, and mobile broadband coverage — split at the median into low-resource and higher-resource wards. This is consistent with the EdLab other studies, which stratify on the basis of remoteness and/or mobile internet connectivity, thus, confirming broadband access as a first-order source of heterogeneity in the Tanzanian context. The test asks: does the treatment effect on module completion differ significantly between low-resource and higher-resource wards? A positive and significant interaction — meaning the intervention raises engagement more in higher-resource wards — would be evidence of inequality amplification. A null or negative interaction would suggest the intervention is equitable or equity-improving. School-level inputs (e.g., number of registered teachers and enrolment) are first aggregated to the ward level, then combined with ward-level indicators

(implementation phase and broadband coverage) to construct a composite index. The index is standardised and split at the median to classify wards as low-resource or higher-resource.

Performance equity interaction: Treatment is interacted with the school's baseline eLMS engagement quartile, derived from Matomo data in the term prior to randomization. We specifically examine the catch-up effect for schools in the bottom quartile of pre-intervention module completion. A positive interaction coefficient for this group — indicating larger treatment effects for the previously lowest-engaging schools — would suggest the intervention is equity-improving along the performance dimension. A negative interaction would indicate that the intervention primarily activates schools that were already somewhat engaged, leaving the most disengaged schools behind. This test directly addresses whether the ward-level visibility mechanism reaches schools that most need improvement, or whether it merely accelerates progress in schools already on an upward trajectory.

This dual framing — resource equity and performance equity — treats distributional consequences as first-order scientific questions. Together they allow the study to distinguish four possible outcomes: an intervention that is equitable on both dimensions; one that is equitable on resources but regressive on performance (or vice versa); and one that amplifies existing inequalities on both dimensions. Each combination carries distinct policy implications for how TIE should design any scale-up of the recognition mechanism.

2.3.4 Threats to identification and mitigations

Threat	Description	Mitigation
Spillover between wards	WEOs in control wards may learn of the intervention from WEOs in treated wards and informally adopt similar practices.	Randomization is stratified within districts, ensuring treatment arms are balanced geographically. Spillovers are expected to be limited due to the operational independence of WEOs. Contamination will be measured through an endline WEO survey and accounted for in robustness checks.
WEO non-compliance (T1/T2)	Some WEOs may not open or act on the report, attenuating the treatment effect.	SMS delivery receipts track report opening. Non-compliance is accounted for in the LATE analysis. The ITT remains unbiased regardless of compliance.
Headteacher gaming	Headteachers in treated schools may pressure only certain teachers to log in, inflating the completion metric without genuine engagement.	Module completion (not just login) is the primary outcome, requiring substantive platform engagement. Post-intervention survey measures self-reported coercion and distinguishes voluntary from compliance-driven engagement.
Differential attrition	Teacher transfers or school closures may disproportionately affect treated or control schools.	Attrition rates will be compared across arms at endline. Analysis will be conducted on the intention-to-treat sample (all schools at baseline) with sensitivity checks excluding schools with high mid-study attrition.

Please note that the intervention does not expose teachers to sanctions or punitive consequences. The ward-level report is positive in framing, names top-performing schools, and does not publicly identify low-performing schools by name in the version shared with teachers. WEOs in T2 are prompted to hold a supportive conversation, not to issue warnings. Ethical approval will be sought from Georgetown University's IRB prior to implementation.

2.3.5 Balance and Randomization Checks

To verify the internal validity of the cluster randomization, we will conduct balance tests using baseline administrative data prior to the intervention launch. Because randomization is stratified, we expect balance by design within strata — the balance checks serve primarily as a verification step and to detect any implementation errors in the randomization procedure.

We will compare all three arms (C, T1, T2) across the following pre-specified baseline covariates:

- Ward-level eLMS module completion rate (from Matomo, term prior to randomization)
- Ward-level eLMS registration rate (share of teachers registered on the platform)
- Average school enrolment per ward (from TIE/EMIS records)
- Number of registered eLMS teachers per ward (from TIE records)
- MEWAKA implementation phase (early vs. later LGA cohort)
- Mobile broadband coverage index (from GSMA/TCRA data)
- CoL session attendance rate (from eLMS attendance registers, term prior to randomization)

For each covariate, we will report means and standard deviations by arm, pairwise differences across arms with associated p-values, and a joint F-test of the null hypothesis that the set of covariates does not predict treatment assignment. We report balance across all three pairwise comparisons — T1 vs. C, T2 vs. C, and T1 vs. T2 — rather than pooling treatment arms, since the T1 vs. T2 comparison is itself a hypothesis of interest.

Importantly, balance is assessed on the full composite resource index used for stratification, not just its individual components. A failure of balance on the composite — which combines enrolment, teacher count, implementation phase, and broadband coverage — would indicate a stratification error and would require a re-examination of the randomization procedure before the intervention launches.

We will not condition the analysis on balance test outcomes. If minor imbalances are detected despite randomization, we address them through covariate adjustment in the regression specification (Section 3.3.1) rather than through re-randomization or post-hoc re-weighting.

3. Data

3.1 Data Sources

3.1.1 TIE Administrative Records

TIE maintains school-level administrative records covering all MEWAKA-registered schools, including district and ward identifiers, number of registered teachers per school, total school enrolment, MEWAKA implementation phase (LGA cohort), and historical eLMS login and completion data by school and term. These records are used for four purposes in this study: constructing the sampling frame of eligible wards (Section 2.2.2); constructing the composite resource index used for stratification, which draws on registered teacher count, school enrolment, and implementation phase (Section 2.2.4); stratifying wards prior to randomization alongside Matomo engagement data and mobile broadband coverage; and providing baseline covariates for the regression analysis and balance checks (Section 2.3.5).

3.1.2 Mobile Broadband Coverage Data

Ward-level mobile broadband coverage is sourced from GSMA Intelligence or, where available, from Tanzania Communications Regulatory Authority (TCRA) national coverage maps. This is the fourth component of the composite resource index (Section 2.2.4) and is not available in TIE administrative records. It is treated as a separate data source because it requires a distinct data acquisition step prior to randomization. Coverage data are merged to the ward sampling frame using ward geographic identifiers. This data source is consistent with the EdLab other studies on mobile internet connectivity, which have identified broadband access as a primary source of heterogeneity in MEWAKA implementation across Tanzania.

3.1.3 eLMS Platform Analytics (Matomo)

The eLMS platform logs all user activity through Matomo, an open-source web analytics system. Matomo records each login event, the module accessed, time on platform, and — crucially for this study — whether a module assessment was completed. All data are linked to individual teacher accounts by a unique teacher ID, which can be aggregated to school and ward level. This is the primary outcome data source. Data are already collected continuously and accessible to the EdLab research team through TIE's analytics dashboard. No additional data collection burden is imposed on schools or teachers. Pre-intervention Matomo data also provide the baseline completion rate used for stratification and the empirical standard deviation needed to finalise the MDE calculation in Section 4.

3.1.4 CoL Session Attendance Registers

Teachers submit CoL session attendance registers through the eLMS platform after each session. These registers are maintained continuously by TIE and are accessible to the EdLab research team under the existing data sharing agreement. The CoL attendance data allow this study to distinguish between two related but distinct behaviors: self-directed eLMS module engagement (captured by Matomo) and participation in the weekly school-based Communities of Learning (captured by attendance registers). A school could respond to the ward-level visibility intervention by increasing CoL attendance without increasing independent module completion, or vice versa.

3.1.5 Post-Intervention Teacher Survey

A brief structured survey is administered to a random sample of teachers within treated and control schools at endline, approximately six months after intervention commencement. The survey serves two purposes: validating and contextualising the administrative outcomes — distinguishing voluntary engagement from headteacher-directed compliance, and measuring teacher awareness of the ward ranking report; and characterising the teacher-level predictors of uptake for the third research question, including gender, years of experience, subject area, and self-reported digital confidence. The survey is conducted by phone using SurveyCTO (CAPI/CATI) and is estimated at 20 minutes. A random sample of 10 teachers per treated school and 5 per control school is targeted, yielding approximately 4,500 teacher-level observations for the heterogeneity analysis. A short module will also be administered to headteachers in sampled schools to measure awareness of the ward-level ranking and any resulting school-level responses.

3.2 Outcomes of Interest

3.2.1 Inputs

The inputs are the specific resources and activities provided by the study team and TIE. These will be tracked to ensure treatment fidelity:

- **Ward-Level Engagement Report (T1 and T2):** Monthly one-page PDF ranking all schools in the ward by module completion rate, generated automatically from Matomo analytics and delivered to the WEO via SMS link.
- **Accountability Prompt (T2 only):** A structured written prompt delivered alongside the report, asking the WEO to discuss findings with the bottom two schools during their next scheduled visit and record a brief note on the conversation.
- **SMS Delivery Tracking:** Confirmation receipts and link-open logs for each report delivery, used to verify WEO exposure for the LATE analysis.

3.2.2. Outputs

These metrics indicate whether the report successfully reached and activated the WEO-headteacher accountability channel:

- **WEO Report Opening Rate:** Share of T1 and T2 WEOs who opened the SMS report link each month, confirmed via link-tracking.
- **Headteacher Awareness Rate:** Share of headteachers in treated schools who can accurately report their school's ward ranking position at endline (from post-intervention survey).
- **School-Level Login Activity:** Number of unique teacher logins per school in the week following each report delivery, as an early leading indicator of behavioral response prior to the primary outcome window.

3.2.3 Primary outcomes

All primary outcomes are constructed from Matomo administrative data at the school level, aggregated across the six-month intervention period.

Outcome	Construction	Source
Module completion rate	Share of registered teachers in the school who completed at least one eLMS module assessment during the intervention period. This is the primary metric used in the ward-level ranking report.	Matomo (teacher-level, aggregated to school)
Sustained engagement rate	Share of registered teachers with at least two distinct login sessions separated by 14 or more days. Captures whether engagement is sustained or a one-off response to the report.	Matomo (teacher-level login timestamps)
Within-school completion concentration (equity metric)	Herfindahl-Hirschman Index of module completions across teachers within a school. A score approaching 1 indicates one teacher completing all modules; a low score indicates broad distribution. Captures whether treatment induces gaming by concentrating completions among a few teachers.	Matomo (teacher-level completions)

3.2.4 Secondary outcomes

Outcome	Construction	Source
Average logins per registered teacher	Total login events divided by number of registered teachers in the school.	Matomo
CoL session attendance rate	Share of teachers attending at least one CoL session per month, from registers submitted through the eLMS. Distinguishes collective CoL participation from self-directed module completion, allowing a test of whether the intervention shifts one behavior without the other.	eLMS CoL attendance registers
Module completion gap: high-resource vs. low-resource	Difference in school-level completion rates between low-resource and high-resource schools within the same treatment arm. Pre-specified equity heterogeneity measure.	Matomo + TIE admin records
WEO report engagement	Whether the WEO opened the report link (confirmed via SMS delivery receipt and link-tracking). Used in LATE analysis.	SMS delivery logs
Headteacher awareness of ranking	Whether the headteacher can accurately report their school's position in the ward ranking at endline.	Post-intervention survey
Self-reported coercion	Whether teachers report being asked by their headteacher to log in specifically because of the ranking. Identifies gaming responses.	Post-intervention survey

3.3 Hypotheses and Outcome Construction

The following hypotheses are pre-registered in the order stated. No hypotheses will be added after randomization.

#	Hypothesis	Primary outcome variable	Test
H1	T1 and T2 schools achieve higher module completion rates than control schools at endline.	School-level module completion rate	OLS with stratification cell fixed effects and baseline completion rate as a covariate. ITT estimate.
H2	T2 schools (visibility + accountability prompt) achieve higher completion rates than T1 schools (visibility only).	School-level module completion rate	Direct comparison T2 vs T1. Tests whether the WEO accountability prompt adds effect beyond visibility alone.
H3 (equity)	The treatment effect on module completion does not differ significantly between low-resource and high-resource schools.	Interaction: treatment × resource stratum	OLS interaction term. Null hypothesis is equity. A positive and significant interaction (high resource schools gain more) is evidence of inequality amplification.

3.3.1 Econometric Specification

To evaluate the primary, secondary, and equity-based impacts of the intervention, we estimate the following regression model for each outcome:

$$Y_{is} = \beta_0 + \beta_1 T1_s + \beta_2 T2_s + \beta_3 High_Res_s + \beta_4 (T1_s \times High_Res_s) + \beta_5 (T2_s \times High_Res_s) + \beta_6 \bar{Y}_{s0} + \gamma_s + \varepsilon_{is}$$

where:

- Y_{is} is the eLMS module completion rate for school *i* in ward *s*, measured over the six-month intervention period
- $T1_s$ is a binary indicator equal to 1 if ward *s* was assigned to the visibility-only arm (monthly WEO report, no accountability prompt)
- $T2_s$ is a binary indicator equal to 1 if ward *s* was assigned to the visibility plus accountability arm. Control wards are the omitted category
- $High_Res_s$ is a binary indicator equal to 1 if ward *s* is classified as high-resource based on TIE administrative data and the composite resource index described above.
- $T1_s \times High_Res_s$ and $T2_s \times High_Res_s$ are interaction terms between each treatment arm and resource status
- $\bar{Y}_{s0}$ is the ward-level baseline module completion rate, derived from Matomo data in the term prior to randomization
- γ_s are stratification cell fixed effects (one per stratum: high-resource/above-median, high-resource/below-median, low-resource/above-median, low-resource/below-median), absorbing all time-invariant differences across strata
- ε_{is} is the error term, clustered at the ward level to account for the group-randomized design

Parameters of interest and hypothesis mapping:

Coefficient	Interpretation	Hypothesis
β_1	ITT effect of T1 (visibility) relative to control for **low-resource wards**	H1
β_2	ITT effect of T2 (visibility + accountability) relative to control for **low-resource wards**	H1
$\beta_2 - \beta_1$	Additional effect of accountability (T2 vs. T1)	H2
β_4	Additional effect of T1 for **high-resource wards relative to low-resource wards**	H3
β_5	Additional effect of T2 for **high-resource wards relative to low-resource wards**	H3

All estimates are intent-to-treat (ITT). Standard errors are clustered at the ward level.

Equity interpretation: A positive and significant β_4 or β_5 means high-resource schools respond more strongly to the intervention than low-resource schools. Because high-resource schools already have higher baseline engagement, a positive interaction is inequality-amplifying — it widens the existing resource gap. A negative interaction means the intervention disproportionately benefits low-resource schools, which would be equity-improving. The null hypothesis for H3 is $\beta_4 = \beta_5 = 0$ (no differential effect by resource level).

LATE specification: To estimate the effect of actual WEO engagement with the report (rather than assignment to treatment), we estimate a two-stage least squares model instrumenting WEO report compliance — defined as confirmed link-opening via SMS delivery tracking — with treatment arm assignment. The first stage regresses WEO compliance on T1 and T2; the second stage uses predicted compliance to estimate the effect among wards where the WEO actually opened the report. This addresses the concern that non-compliant WEOs attenuate the ITT estimate and allows a cleaner test of the visibility mechanism.

3.4 Data Collection Strategy and Timeline

Phase	Activity	Timing
Baseline	Extract school-level eLMS records from TIE admin data for stratification and baseline covariate construction. Compile WEO contact list.	Month 0 (pre-randomization)
Randomization	Assign wards to C / T1 / T2 using pre-registered procedure. Lock randomization list.	Month 0
Intervention	Monthly ward-level engagement reports generated from Matomo and delivered to WEOs in T1 and T2 wards. T2 WEOs receive structured discussion prompt.	Months 1–6
Continuous monitoring	Matomo data and CoL attendance registers extracted monthly. SMS delivery receipts logged. No data shared with control wards.	Months 1–6
Endline survey	Phone survey administered to random sample of 10 teachers per treated school and 5 per control school.	Month 6–7
Analysis	Primary ITT regression and heterogeneity analysis. LATE analysis using WEO report opening as instrument.	Month 7–8

Note on data access: Matomo analytics data, TIE administrative records, and CoL attendance registers are accessible to the EdLab research team under an existing data sharing agreement between gui2de Georgetown and TIE. No new data access agreements are required. The post-intervention survey protocol will be submitted for IRB review at Georgetown University prior to implementation. Note on learning outcomes: SFNA and PSLE administrative exam data exist and are used by the EdLab's other studies to measure student learning outcomes at scale. This study does not use exam data because the causal chain from eLMS engagement to measurable learning gains requires a longer study window than the six-month intervention period. eLMS module completion is the proximal outcome this study is powered to detect; student learning outcomes are the distal motivation articulated in the introduction.

4. Power Calculations

4.1 Setup and Parameters

All power calculations assume 80 percent power and a 5 percent two-sided significance level ($M = z_{0.025} + z_{0.2} \approx 2.80$). The study is a three-arm cluster-randomized trial with wards as the unit of randomization and schools as the unit of observation. The primary outcome is the school-level module completion rate: the share of registered

teachers in a school who completed at least one eLMS module assessment during the six-month intervention period.

The MDE formula for a two-arm pairwise comparison in a cluster RCT is:

$$MDE = M \times \sqrt{[\ (2 \times (1 - R^2) \times DEFF) \ / \ (J \times n)\ ]}$$

where J is the number of wards per arm, n is the number of schools per ward, R^2 is the variance explained by the baseline covariate (lagged school-level completion rate), and DEFF = 1 + (n − 1) × ICC is the design effect. Three pairwise comparisons are conducted: T1 vs. C, T2 vs. C, and T2 vs. T1.

4.2 Parameter Assumptions

Parameter	Value	Source / Justification
Wards per arm (J)	40	Determined by feasibility and budget. Target total of 120 wards across three arms.
Schools per ward (n)	5	Consistent with EdLab CoL study administrative data (~4–6 schools per ward in MEWAKA districts). Conservative mid-point used.
ICC (ward-level)	0.10	Ward-level ICC for school-level outcomes. The EdLab PAP (Cilliers, Habyarimana, and Lucas, 2026) estimates student-level ICC of 0.25–0.29 from SFNA data. School-level ward clustering will be lower; ICC = 0.10 is a conservative mid-point. Sensitivity analysis spans 0.05–0.15.
R^2 (baseline covariate)	0.50	Variance explained by lagged ward-level eLMS completion rate as a covariate. Consistent with the EdLab PAP's conservative R^2 = 0.50 (observed adj. R^2 = 0.57 in SFNA regressions with district fixed effects and lagged scores).
Baseline completion rate	~2%	From TIE eLMS administrative data: fewer than 2 percent of registered teachers have completed a module assessment (Tanzania Institute of Education, 2025).
SD of school-level completion rate	0.10	Assumed. Near-zero baseline with high between-school variance expected. Sensitivity analysis uses SD = 0.08–0.12.

4.3 Minimum Detectable Effects

Under the preferred specification (ICC = 0.10, R^2 = 0.50, J = 40 wards per arm, n = 5 schools per ward), the MDE for each pairwise comparison is 0.234 standard deviations of the school-level completion rate. Translating to percentage points using the assumed SD of 0.10, this corresponds to approximately 2.3 percentage points.

Policy relevance check: A 2.3 percentage point MDE represents a doubling of the baseline completion rate (from approximately 2 percent to 4.3 percent). This is ambitious given the near-zero baseline, but plausible: the BRN school ranking reform raised learning outcomes by approximately 0.25 standard deviations (Cilliers, Mbiti, and Zeitlin, 2021), and the Habyarimana and Cilliers (2024) WEO text message intervention improved teaching

practices by 0.27 standard deviations. Both are informationally similar interventions operating through the same bureaucratic reputational channel. The MDE of 0.234 SD is calibrated to detect effects of this magnitude, providing reasonable confidence that a genuine effect will not be missed. The following table presents MDEs across the sensitivity analysis ranges.

Table X: MDE Sensitivity Analysis (J = 40 wards per arm, n = 5 schools per ward)

ICC	$R^2 = 0.30$	$R^2 = 0.50$	$R^2 = 0.30$ (pp, σ=0.10)	$R^2 = 0.50$ (pp, σ=0.10)
0.05	0.257 SD	0.217 SD	2.6 pp	2.2 pp
0.10	0.277 SD	0.234 SD	2.8 pp	2.3 pp
0.15	0.297 SD	0.251 SD	3.0 pp	2.5 pp

Table X: MDE by Sample Size (ICC = 0.10, $R^2 = 0.50$, n = 5)

Wards per arm (J)	Total wards	MDE (SD)	MDE (pp, σ=0.10)
30	90	0.271 SD	2.7 pp
40	120	0.234 SD	2.3 pp ← preferred
60	180	0.191 SD	1.9 pp
80	240	0.166 SD	1.7 pp

4.4 Limitations

The power calculations rest on three assumptions that cannot be verified prior to data collection and that the research team acknowledges honestly.

- **ICC.** No existing estimate of the ward-level ICC for school-level eLMS completion rates exists. The EdLab other studies use an ICC of 0.25–0.29 applies to student-level exam outcomes within schools — a different level of clustering and a different outcome. The ICC of 0.10 used here is an informed assumption. If the true ICC is higher (e.g., 0.20 because wards differ substantially in WEO quality or connectivity), the study will be underpowered. A pilot or baseline data extraction should be used to update this assumption before finalising the sample size.
- **SD of the outcome.** With a near-zero baseline completion rate (~2%), the distribution of school-level completion rates is highly skewed and the SD is difficult to estimate without observed data. The assumed SD of 0.10 may over- or understate the true variance. TIE's Matomo data for the pre-intervention period should be used to compute the empirical SD of school-level completion rates before the study launches, and the MDE should be recalculated accordingly.
- **Effect size expectations.** The BRN and Habyarimana-Cilliers benchmarks are drawn from different outcomes (exam scores, teaching practices) and different interventions. eLMS engagement is a more proximal and lower-stakes behavior than either, which could mean effects are larger (lower baseline, more room to move) or smaller (the outcome matters less to headteachers than exam pass rates). The

study is powered to detect effects of approximately 0.23 SD. Smaller effects — for example, 0.10–0.15 SD — would require 150–250 wards per arm and are beyond the feasible scope of this proposal.

Note: Power calculations were conducted in Python using the standard cluster-RCT MDE formula. Parameters are drawn from the EdLab Tanzania where available, and from TIE administrative data otherwise. All assumptions will be revisited once pre-intervention Matomo data are available for the eligible ward sample.